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1 Sobolev Spaces

Sobolev spaces are defined by norms that measure both smoothness and integrability. Given a domain $\Omega \subseteq \mathbb{R}^d$, we define

$$\|f\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^p}^p \right)^{1/p},$$

with suitable adjustment for $p = \infty$. Though note that this is just the ℓ^p norm, so it could be replaced by any other norm up to loss of constant. Consequently you might also see, for example,

$$\|f\|_{W^{k,p}(\Omega)} := \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^p},$$

and this is the same. The important thing is that it controls the L^p norm of every derivative up to order k at once. With any of these equivalent norms, $W^{k,p}$ is a Banach space. The most important of these are $H^k = W^{k,2}$, which are Hilbert spaces with inner product

$$\langle f, g \rangle_{H^k} = \sum_{|\alpha| \leq k} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2}.$$

The smooth functions that lie in $W^{k,p}$ form a dense subset. For $\Omega = \mathbb{R}^d$, in fact the set of compactly supported smooth functions is dense, but this is not true any other subdomain. This closure (denoted $W_0^{k,p}$) is a sub-Banach space, and it is important for Dirichlet problems.

Perhaps the most important theorem of Sobolev spaces is the so-called Sobolev embedding theorem. Intuitively, we can “trade” derivatives to gain integrability. If you have a lot of derivatives, then you find that the function just actually be classical. Note: the standard Sobolev embeddings are for $W_0^{k,p}$. We will work on $\mathbb{R}^n$, where this is just $W^{k,p}$. NOTE: $p < \infty$.

- If $k \leq d/p$, then $W^{j+k,p} \hookrightarrow W^{j,q}$ for $q = \frac{dp}{d-kp}$. In particular, $W^{k,p} \hookrightarrow L^q$. (And you can get lower q up to p , down to 1 if finite measure I assume.) [TODO: Check the endpoints]
- If $k > d/p$, then $W^{k,p} \hookrightarrow C^{r,\alpha}$, where $r + \alpha = d/p - k$.

Below q we have compact embeddings. This is called the Rellich–Kondrachev theorem.

2 Sobolev Spaces on Manifolds

For a smooth Riemannian manifold (M, g) , an integer $k \geq 0$, and a function $u \in C^\infty(M, \mathbb{R})$, we can define the k th covariant derivative $\nabla^k f$ (the Levi–Civita connection iterated k times), which is a k -covariant tensor. Examples: $\nabla f = df$, $\nabla^2 f$ is the Hessian tensor. In a chart, we can define the norm

$$|\nabla^k u|^2 = g^{i_1 j_1} \dots g^{i_k j_k} (\nabla^k u)_{i_1 \dots i_k} (\nabla^k u)_{j_1 \dots j_k}.$$

Thus, for example, in $\mathbb{R}^d$,

$$|\nabla u| = \left(\sum_{i=1}^d |\partial_i u|^2 \right)^{1/2},$$

as we would expect. For $p \in [1, \infty)$ (Hebey doesn't address $p = \infty$), we can define

$$\|u\|_{W^{k,p}} = \sum_{j=0}^k \left(\int_M |\nabla^j u|^p dV_g \right)^{1/p}.$$

This is a norm, and the closure of the set of all C^∞ functions for which this is finite (which is just all of C^∞ if M is compact) will be the Sobolev space $W^{p,k}(M)$. Since the $W^{k,p}$ is a stronger norm than the L^p norm, we see that elements of $W^{k,p}$ are bona fide functions, specifically in L^p .

For $p \in (1, \infty)$, the spaces $W^{p,k}$ are reflexive. (We cannot expect this to hold for $p = 1$, as L^1 is not reflexive.) In fact, the space $W^{k,2}$ is a Hilbert space: the inner product

$$\langle u, v \rangle = \sum_{j=0}^k \int_M \langle \nabla^j u, \nabla^j v \rangle dV_g$$

(this is the inner product on covariant tensor fields induced by the metric). This does not induce the $W^{k,2}$ norm defined above, but it does induce an equivalent norm:

$$\|u\| = \left(\sum_{j=0}^k \int_M |\nabla^j u|^2 dV_g \right)^{1/2}.$$

In general, it is not uncommon to see various different ℓ^p norms used interchangeably, which is justified by the fact that these norms are all equivalent on $\mathbb{R}^N$.

It would be useful if the set of compactly supported smooth functions were also dense in $W^{k,p}$. This is true for $\mathbb{R}^d$, but is not true in general; it is not even true in bounded domains of $\mathbb{R}^d$. However, it is true for complete manifolds if $k = 1$, or if the manifold has positive injectivity radius and the derivatives of order $\leq k - 2$ of its Ricci curvature are bounded. (In particular, it is always true for compact manifolds.)

The theme of Sobolev embedding is that we can “trade” smoothness to get a higher degree of integrability. The basic theorem takes the following form, though note *it does not always hold*: if $0 \leq \ell \leq k$ and $(k - \ell)p < d$, then

$$W^{k,p} \hookrightarrow W^{\ell,q}, \quad \frac{1}{q} = \frac{1}{p} - \frac{k - \ell}{d} \iff q = \frac{dp}{d - (k - \ell)p} \geq p.$$

We can state two important cases for which this holds: Lipschitz domains in $\mathbb{R}^n$ (including $\mathbb{R}^n$ itself, and compact manifolds).

A particularly important case is seeing how Sobolev spaces embed into L^q spaces, i.e., $\ell = 0$. In this case, the Sobolev embedding is that if $kp < d$, then

$$W^{k,p} \hookrightarrow L^q, \quad q = \frac{dp}{d - kp}.$$

For example, on $\mathbb{R}^4$ or a compact 4-manifold, we have the following Sobolev inequality:

$$\|u\|_{L^4} \leq C\|u\|_{W^{1,2}} \leq C\|u\|_{W^{2,2}},$$

which is an inequality I got from Donaldson–Kronheimer.

There is also a second part of the Sobolev embedding: if $(k - \ell)p > d$, i.e., if we have a great deal of “weak” derivatives, then we actually have classical derivatives. Namely,

$$W^{k,p} \hookrightarrow C_B^\ell = \left\{ u : \|u\|_{C^\ell} = \sum_{j=0}^\ell \sup_M |\nabla^j u| < \infty \right\}.$$

There are more refined estimates one can get in terms of Hölder spaces, sometimes called Sobolev or Morrey estimates, but I assume these are less relevant. These inequalities again do not hold for every manifold, but do hold for Lipschitz domains in $\mathbb{R}^n$ and compact manifolds.

The middle, where $(k - \ell)p = d$, the so-called critical case, is not nice. We know, for example, that $W^{1,2} = H^1(\mathbb{R}^2)$, is not included in $L^\infty(\mathbb{R}^2)$.

Anyway, let's look at when we know whether Sobolev embeddings hold, starting with the first kind (Sobolev into Sobolev/ L^p). A class in which they are known to fail are complete non-compact manifolds ($d \geq 2$) of finite volume. Classes in which they do hold:

- $\mathbb{R}^n$
- compact manifolds
- complete manifolds with Ricci curvature bounded from below and positive injectivity radius
- complete manifolds with Ricci curvature bounded from below and volumes of unit balls bounded from below.

The other type of Sobolev embeddings also hold for complete manifolds with Ricci curvature bounded from below and positive injectivity radius.

For compact manifolds and $\mathbb{R}^n$, we can say even more than just the Sobolev embedding. Recall that we have $W^{k,p} \hookrightarrow W^{\ell,q}$ for

$$q = \frac{dp}{d - (k - \ell)p}.$$

This actually implies an embedding into all lower q as well, up to p , since $W^{k,p} \hookrightarrow L^p$ and we can just interpolate. But for these lower q , the embedding is not just continuous, but compact. This is called the Rellich–Kondrachov theorem, and is useful for certain variational arguments. Unfortunately this is often not true in general.

3 Implicit and Inverse Function Theorems

One can generalize a great deal of differential calculus to maps between Banach spaces; the primary thing that doesn't transfer is representing a derivative using

partial derivatives. Suppose we have a map of Banach spaces $f : X \rightarrow Y$. We say that f is (Fréchet) differentiable at a point $x \in X$ if there exists a bounded linear map $Df_x : X \rightarrow Y$ such that

$$f(x+h) - f(x) - Df_x(h) = o(\|h\|).$$

Theorem 1. *Let f be C^1 around x_0 and Df_{x_0} be an isomorphism. Then there exists a neighborhood U of x_0 and a neighborhood V of $f(x_0)$ such that $f : U \rightarrow V$ is a diffeomorphism. (At all points, f is invertible and its inverse is C^1 .) Moreover,*

$$D(f^{-1})_{f(x)} = Df_x^{-1}.$$

Theorem 2. *Let $f : Z \times X \rightarrow Y$ be a C^1 function around (z_0, x_0) . There exist open neighborhoods W of z_0 , U of x_0 , and V of $f(z_0, x_0)$, and a unique map $g : W \times V \rightarrow U$, such that*

$$f(z, g(z, y)) = y, \quad \forall (z, y) \in W \times V.$$

This map is C^1 and its derivative is given by

$$Dg(z, y)_{(\eta, \zeta)} = \partial_x f_{(z, x)}^{-1}(\zeta - \partial_z f_{(z, x)}\eta).$$

4 Elliptic Operators

As with most things in PDEs, the definition of an elliptic operator is fairly loose, but the one thing that is certain is that the Laplacian is the prototypical example. The prototypical elliptic PDE is the Laplace equation $\Delta u = 0$, or its inhomogeneous form, the Poisson equation $\Delta u = f$. The same is true in differential geometry, to some extent: the prototypical elliptic operator is the Laplace–Beltrami operator.

(Nicolaescu, 10.1) Given two bundles over a manifold, we can define partial differential operators. For any $x_0 \in M$, and $\xi_1, \dots, \xi_m \in T_{x_0}^*M$, an m th order PDO P induces a linear map

$$\sigma(P)(\xi_1, \dots, \xi_m) : E_{x_0} \rightarrow F_{x_0}.$$

This defines the symbol of P . In local coordinates, if we view P as

$$\sum_{|\alpha| \leq m} A_\alpha(x) \partial^\alpha, \quad \xi = \xi_i dx^i,$$

then the symbol is the polynomial

$$\sigma(P)(\xi) = \sum_{|\alpha| \leq m} A_\alpha(x) \xi^\alpha.$$

(See this question¹.) The highest homogeneous part is called the principal symbol, $\sigma_m(P)$. A PDO is called elliptic if, for any $x \in M$ and $\xi \in T_x^*M \setminus \{0\}$, the principal symbol $\sigma_m(\xi) : E_x \rightarrow F_x$ is an isomorphism.

In general, the Laplacians of differential geometry are standard examples of elliptic PDOs, including the Laplace–Beltrami operator and the Hodge Laplacian.

5 The Calderón–Zygmund Inequality

The Calderón–Zygmund inequality states that

$$\|\nabla^2 u\|_{L^p} \lesssim_{d,p} \|u\|_{L^p} + \|\Delta u\|_{L^p}, \quad u \in \mathcal{D}(\mathbb{R}^d), 1 < p < \infty.$$

In fact, in the $\mathbb{R}^d$ case, we even have

$$\|\nabla^2 u\|_{L^p} \lesssim_{d,p} \|\Delta u\|_{L^p}, \quad u \in \mathcal{D}(\mathbb{R}^d), 1 < p < \infty,$$

which follows from singular integral theory. (The endpoints are not true.) However, I do not think this stronger form holds in general on manifolds, so we will ignore it. Usually, the way we see this inequality in action is that we have a solution to a Poisson problem, $\Delta u = f \in L^p$, and we can deduce that any L^p solution is in $W^{2,p}$.

This inequality remains true on compact manifolds, with Δ in this case standing for the Laplace–Beltrami operator. More generally, it holds for all p on complete manifolds whose Ricci curvature is bounded (above and below) and whose injectivity radius is finite. But it does not hold in general². It is true that the L^2 version holds if we just have a lower bound on the Ricci curvature, as a consequence of the Bochner identity³.

However, we do have local, interior forms of this inequality as well: if we work in an open set Ω , then it holds true when we take the L^p bound of the Hessian in a compact subdomain. Consequently, local forms should be expected to hold in any manifold, regardless of curvature.

¹<https://math.stackexchange.com/questions/589522/definition-of-the-principal-symbol-of-a-differential-operator-on-a-real-vector-b>

²<https://arxiv.org/pdf/2011.13025>

³<https://arxiv.org/pdf/2011.03220>

6 Elliptic Regularity

The signature property of elliptic operators is their regularity property. The Laplacian has the following property: the singular support of Δu is no larger than the singular support of u . In particular, if f is smooth, then any distributional solution to $\Delta u = f$ is also smooth. Fittingly, this property is called hypoellipticity.

At lower levels of regularity, we expect that solutions to elliptic equations of second order (such as the Poisson equation) will have “two more derivatives” than the input data, the coefficients and forcing. More generally, an m th order elliptic PDE should produce solutions that have m more derivatives. For example, on $\mathbb{R}^n$, if L is a C^∞ m th order elliptic operator, then $u \in L^p_{\text{loc}}$ and $Lu \in L^p_{\text{loc}}$ implies $u \in W^{m,p}_{\text{loc}}$, $p \in (1, \infty)$ (Nicolaescu Theorem 10.3.6). More generally, if $u \in L^p_{\text{loc}}$ and $Lu \in W^{k,p}_{\text{loc}}$, then $u \in W^{k+m,p}_{\text{loc}}$. We can see this as intimately connected to Calderón–Zygmund: all of the m th derivatives are controlled in L^p by an elliptic operator.

Similarly, we can derive estimates for Hölder norms: given Hölder data, one can gain m derivatives with the same $\alpha \in (0, 1)$: Nicolaescu states it (Theorem 10.3.11) as saying that, in the *compact* case if $u \in C^{m,\alpha}$ and $Lu \in C^{k,\alpha}$, then $u \in C^{k+m,\alpha}$. (Presumably local Hölder estimates also hold in general, but he does not appear to write these.) These are often called Schauder estimates.

These can be turned into quantitative local inequalities, though of course we should not expect this to become global unless we are in the compact case.

From this we can derive a general technique known as bootstrapping: if our data is sufficiently regular, we can hope to apply elliptic regularity to lift it to the level of the data. For example, we can immediately derive not only that the Laplacian is hypoelliptic, but that any elliptic operator with smooth coefficients is hypoelliptic as well.

In general, this local elliptic regularity extends to general manifolds by charts, partitions of unity, etc.